

LC-3 Assembly Language Quiz

Answer Key

1. ADD — register mode

```
ADD R2, R3, R4    ; R2 ← R3 + R4
```

2. ADD — immediate mode

```
ADD R5, R5, #-1   ; R5 ← R5 - 1
```

3. AND — register mode

```
AND R2, R3, R4    ; R2 ← R3 AND R4
```

4. AND — immediate mode (clear to zero)

```
AND R0, R0, #0    ; R0 ← 0
```

5. NOT

```
NOT R4, R2        ; R4 ← NOT(R2)
```

6. BRz — branch if zero

```
BRz DONE         ; jump to DONE if last result was zero
```

7. BRnp — branch if not zero

```
BRnp LOOP        ; jump to LOOP if last result was non-zero
```

8. JMP

```
IN               ; R0 ← char
```

9. LD — PC-relative load

```
LD R1, NUM       ; R1 ← mem[ NUM ]
```

10. LDI — indirect load

```
LDI R2, PTR      ; R2 ← mem[ mem[ PTR ] ]
```

11. LDR — base+offset load

```
LDR R0, R5, #3   ; R0 ← mem[ R5 + 3 ]
```

12. LEA + PUTS trap

```
LEA R0, MSG      ; R0 ← address of MSG
PUTS             ; print null-terminated string at R0
```

13. ST — store

```
ST R4, RESULT    ; mem[ RESULT ] ← R4
```

14. STI — indirect store

```
STI R4, ADDR_PTR ; mem[ mem[ ADDR_PTR ] ] ← R4
```

15. STR — base+offset store

```
STR R6, R3, #-2  ; mem[ R3 - 2 ] ← R6
```

16. - LEA and LDR

```
LEA R1, DECIMAL
LDR R0, R1, #0
```

17. - Count loop

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```
LD R0, COUNT
LOOP ADD R0, R0, #-1
BRnp LOOP
```